

Rigorous Convergence Analysis of Federated Learning with Partial Forgetting Compensation (FL-PFC)

Abstract—This paper presents a rigorous convergence analysis for federated learning with partial forgetting compensation (FL-PFC) under three challenges: Non-IID clients, partial participation, and proxy-based knowledge distillation. The analysis proceeds in two steps: first establishing convergence on a fixed surrogate objective, then extending to the actual adaptive objective via a conditional-expectation-freezing technique. Without relying on the decay of dynamic weight bias, only a constant bias upper bound is used, yielding the conclusion that the algorithm converges to an $\mathcal{O}(G^2)$ error neighborhood rather than an exact stationary point. The main body presents core theory with proof sketches, while complete step-by-step derivations appear in the Appendix.

I. CONVERGENCE ANALYSIS

Problem formulation. Consider K clients. In communication round t , a subset S_t of clients is sampled for training. Client k minimizes the local objective

$$\tilde{\mathcal{L}}_t^k(\omega; \omega_t) = \mathcal{L}_C^k(\omega) + \lambda_t^k \mathcal{L}_{\text{KD}}^k(\omega) + \frac{\mu_t}{2} \|\omega - \omega_t\|^2, \quad (1)$$

where $\mathcal{L}_C^k$ denotes the classification loss, $\mathcal{L}_{\text{KD}}^k$ the knowledge distillation loss, and the last term a proximal regularizer. Define two analytical objects: the adaptive round-wise objective $\tilde{F}_t(\omega) \triangleq \sum_k p_k \mathbb{E}_\xi[\tilde{\mathcal{L}}_t^k]$ and the time-uniform reference objective $F(\omega) \triangleq \sum_k p_k F_k(\omega)$ with $p_k = n_k/n$.

Let $g_{t,e}^k$ denote the stochastic gradient, $\bar{g}_t^k \triangleq \frac{1}{E} \sum_{e=0}^{E-1} g_{t,e}^k$, and $g_t \triangleq \sum_k p_k \bar{g}_t^k$ the ideal full-participation gradient. Define the actual aggregated gradient $d_t \triangleq \sum_k \alpha_{k,t} \bar{g}_t^k$, bias $b_t \triangleq d_t - g_t$, and proxy residual $r_t \triangleq \beta_t(\omega_t^{\text{proxy}} - \omega_t)$. Weights $\alpha_{k,t}, \beta_t$ follow dynamic weighting and the proxy schedule $\gamma_t^{\text{proxy}} = \bar{C}_\gamma(t+1)^{-(1+\rho)}$ ($\rho > 0$). Let $\mathcal{F}_t$ denote the σ -field at the start of round t and $\mathbb{E}_t[\cdot] \triangleq \mathbb{E}[\cdot | \mathcal{F}_t]$.

Server update decomposition (key identity).

$$\omega_{t+1} = \sum_{k \in S_t} \alpha_{k,t} \omega_{t+1}^k + \beta_t \omega_t^{\text{proxy}} = \omega_t - \eta_t E(g_t + b_t) + r_t. \quad (2)$$

This identity separates the update into three components: $-\eta_t E g_t$ (ideal gradient step), $-\eta_t E b_t$ (bias from partial participation and dynamic weights), and r_t (proxy residual decaying with t). Define the unified constant

$$c_g \triangleq 4 \max\{1 + 8L^2 \bar{\eta}^2 E^2, M(1 + 8L^2 \bar{\eta}^2 E^2)\}, \quad (3)$$

where $\bar{\eta}$ bounds η_t and $c_g \geq 4M$.

A. Assumptions

Assumption 1 (Smoothness, Unbiasedness, Variance). Each F_k is L -smooth with $F_{\text{inf}} \triangleq \inf_\omega F(\omega) > -\infty$; $\mathbb{E}[g_{t,e}^k | \mathcal{F}_t, \omega_{t,e}^k] = \nabla F_k(\omega_{t,e}^k)$; $\mathbb{E}_t \|g_{t,e}^k - \nabla F_k\|^2 \leq \sigma_t^2$.

Assumption 2 (Heterogeneity and Boundedness). $\exists M \geq 1, \kappa \geq 0$ such that $\sum_k p_k \|\nabla F_k(\omega)\|^2 \leq \kappa^2 + M \|\nabla F(\omega)\|^2$. Adaptive coefficients are bounded ($\lambda_t^k \leq \lambda_{\text{max}}, \mu_t \leq \mu_{\text{max}}$), with states in a compact set. Every round $S_t \neq \emptyset$ and $\pi_k \geq \pi_{\text{min}} > 0$.

Assumption 3 (Algorithmic Bridging). $\|\bar{g}_t^k\| \leq G$ almost surely; $\sup_t \|\omega_t^{\text{proxy}} - \omega_t\| \leq R_\beta$.

Proposition 1 (Compactness and Projection). If a compact set $\mathcal{K}_\omega$ contains all $\omega_t, \omega_{t,e}^k, \omega_t^{\text{proxy}}$ and the stochastic gradient map is continuous, then Assumption ?? holds (with G equal to the maximum gradient norm on the compact set and $R_\beta = \text{diam}(\mathcal{K}_\omega)$). With projection-clipping and Euclidean projection, boundedness conditions in Assumption ?? also hold.

B. Bias and Residual

Proposition 2 (Bias Decomposition). $\|b_t\| \leq 2G$. Decompose $b_t = m_t + \zeta_t$ with $m_t \triangleq \mathbb{E}_t[b_t]$ and $\mathbb{E}_t[\zeta_t] = 0$. By Jensen's inequality, $\|m_t\|^2 \leq 4G^2$ and $\mathbb{E}_t \|\zeta_t\|^2 \leq 4G^2$. Denote $B_\mu^2 \triangleq 4G^2$ and $B_V^2 \triangleq 4G^2$ (constant bounds). Using only this constant bound, convergence is to an $\mathcal{O}(G^2)$ error neighborhood rather than exact stationarity.

Proposition 3 (Proxy Residual). $\|r_t\|^2 \leq D/(t+1)^{2+2\rho}$ with $D \triangleq (\pi_{\text{proxy}} \bar{C}_\gamma R_\beta / \pi_{\text{min}})^2$; hence $\sum_{t=0}^\infty (\frac{8}{\eta_t E} + \frac{3L}{2}) \mathbb{E} \|r_t\|^2 < \infty$.

C. Core Lemmas

Lemma 1 (Client Drift Bound). Define $D_e \triangleq \sum_k p_k \mathbb{E}_t \|\omega_{t,e}^k - \omega_t\|^2$. If $8L^2 \eta_t^2 E^2 \leq 1$, then for all $e \leq E$,

$$D_e \leq 2\eta_t^2 e^2 (2\sigma_t^2 + 4\kappa^2 + 4M \|\nabla F(\omega_t)\|^2). \quad (4)$$

Proof sketch. From $\omega_{t,e}^k - \omega_t = -\eta_t \sum_{i=0}^{e-1} g_{t,i}^k$, Cauchy-Schwarz yields $D_e \leq \eta_t^2 e \sum_{i=0}^{e-1} \sum_k p_k \mathbb{E}_t \|g_{t,i}^k\|^2$. Decompose the stochastic gradient as $g_{t,i}^k = \nabla F_k(\omega_{t,i}^k) + \varepsilon_{t,i}^k$. By Assumption ?? and $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$: $\mathbb{E}_t \|g_{t,i}^k\|^2 \leq 2\sigma_t^2 + 2\|\nabla F_k(\omega_{t,i}^k)\|^2$. Pull $\nabla F_k(\omega_{t,i}^k)$ back to ω_t via L -smoothness: $\|\nabla F_k(\omega_{t,i}^k)\|^2 \leq 2\|\nabla F_k(\omega_t)\|^2 + 2L^2 \|\omega_{t,i}^k - \omega_t\|^2$. Combining, weighting by p_k , and applying the heterogeneity assumption ?? gives $\sum_k p_k \mathbb{E}_t \|g_{t,i}^k\|^2 \leq A_t + 4L^2 D_i$

with $A_t \triangleq 2\sigma_l^2 + 4\kappa^2 + 4M\|\nabla F\|^2$. Substituting back, defining $D_{\max,e}$, and using the condition $8L^2\eta_t^2E^2 \leq 1$ yields $D_{\max,e} \leq 2\eta_t^2e^2A_t$. The full five-step derivation appears in Appendix ??.

Lemma 2 (Gradient Bias and Second Moment). *Define $\delta_t \triangleq \mathbb{E}_t[g_t] - \nabla F(\omega_t)$. Under the same conditions:*

$$\mathbb{E}_t\|\delta_t\|^2 \leq 2L^2\eta_t^2E^2A_t, \quad (5)$$

$$\mathbb{E}_t\|g_t\|^2 \leq c_g(\sigma_l^2 + \kappa^2 + \|\nabla F(\omega_t)\|^2). \quad (6)$$

Proof sketch. $\delta_t = \frac{1}{E} \sum_{e,k} p_k(\nabla F_k(\omega_{t,e}^k) - \nabla F_k(\omega_t))$. By Jensen's inequality and L -smoothness: $\mathbb{E}_t\|\delta_t\|^2 \leq \frac{L^2}{E} \sum_e D_e \leq L^2 D_{\max,E} \leq 2L^2\eta_t^2E^2A_t$. For the second moment: $\mathbb{E}_t\|g_t\|^2 \leq A_t + 4L^2D_{\max,E} \leq (1 + 8L^2\eta_t^2E^2)A_t$. Substituting $A_t = 2\sigma_l^2 + 4\kappa^2 + 4M\|\nabla F\|^2$ and bounding termwise via c_g (Eq. (??)) yields Eq. (??). See Appendix ?? for details.

D. Single-Step Descent and Convergence

Lemma 3 (Single-Step Descent). *If the step-size condition $x_t \triangleq \eta_t EL \leq c_0 \leq \frac{1}{16(1+c_g)}$ holds, then*

$$\begin{aligned} \mathbb{E}_t[F(\omega_{t+1})] &\leq F(\omega_t) - \frac{\eta_t E}{8} \|\nabla F(\omega_t)\|^2 \\ &\quad + c_1\eta_t^2E^2(\sigma_l^2 + \kappa^2 + B_V^2) + c_2\eta_t E \cdot B_\mu^2 \\ &\quad + \left(\frac{8}{\eta_t E} + \frac{3L}{2}\right) \mathbb{E}_t\|r_t\|^2, \end{aligned} \quad (7)$$

with $c_1 \triangleq 8Lc_0 + \frac{3L}{2}c_g + 3L$ and $c_2 \triangleq 2 + 3c_0$.

Proof sketch. From L -smoothness: $F(\omega_{t+1}) \leq F(\omega_t) + \langle \nabla F, \Delta_t \rangle + \frac{L}{2}\|\Delta_t\|^2$, where $\Delta_t = -\eta_t E(g_t + b_t) + r_t$.

Inner product: $\mathbb{E}_t\langle \nabla F, \Delta_t \rangle = -\eta_t E\langle \nabla F, \mathbb{E}_t[g_t] \rangle - \eta_t E\langle \nabla F, m_t \rangle + \mathbb{E}_t\langle \nabla F, r_t \rangle$. With $\mathbb{E}_t[g_t] = \nabla F + \delta_t$, applying Young's inequality termwise:

$$\begin{aligned} \text{(A)} \quad &-\eta_t E\langle \nabla F, \nabla F + \delta_t \rangle \leq -\eta_t E\|\nabla F\|^2 + \frac{\eta_t E}{4}\|\nabla F\|^2 + \eta_t E\|\delta_t\|^2 \\ \text{(B)} \quad &-\eta_t E\langle \nabla F, m_t \rangle \leq \frac{\eta_t E}{4}\|\nabla F\|^2 + \eta_t E\|m_t\|^2, \\ \text{(C)} \quad &\mathbb{E}_t\langle \nabla F, r_t \rangle \leq \frac{\eta_t E}{8}\|\nabla F\|^2 \\ &\quad + \frac{2}{\eta_t E} \mathbb{E}_t\|r_t\|^2. \end{aligned}$$

Summing yields

$$\begin{aligned} \mathbb{E}_t\langle \nabla F, \Delta_t \rangle &\leq -\frac{3}{8}\eta_t E\|\nabla F\|^2 \\ &\quad + \eta_t E\|\delta_t\|^2 \\ &\quad + \eta_t E \cdot B_\mu^2 + \frac{2}{\eta_t E} \mathbb{E}_t\|r_t\|^2. \end{aligned}$$

Quadratic term: $\|\Delta_t\|^2 \leq 3\eta_t^2E^2\|g_t\|^2 + 3\eta_t^2E^2\|b_t\|^2 + 3\|r_t\|^2$. Substituting the bounds from Lemma ??: $\frac{L}{2}\mathbb{E}_t\|\Delta_t\|^2 \leq \frac{3L}{2}\eta_t^2E^2c_g(\sigma_l^2 + \kappa^2 + \|\nabla F\|^2) + \frac{3L}{2}\eta_t^2E^2(B_\mu^2 + B_V^2) + \frac{3L}{2}\mathbb{E}_t\|r_t\|^2$.

Merging and coefficient absorption: Substituting $\eta_t E\|\delta_t\|^2 \leq 8L^2\eta_t^3E^3(\sigma_l^2 + \kappa^2 + M\|\nabla F\|^2)$, the coefficient of $\|\nabla F\|^2$ becomes $-\frac{3}{8}\eta_t E + \frac{3L}{2}\eta_t^2E^2c_g + 8L^2\eta_t^3E^3M$. Under $\eta_t EL \leq c_0$, this is $\leq -\frac{\eta_t E}{8}$. Reducing the σ_l^2, κ^2 terms via $8L^2\eta_t^3E^3 \leq 8Lc_0\eta_t^2E^2$ yields $c_1\eta_t^2E^2(\sigma_l^2 + \kappa^2 + B_V^2)$ with $c_1 = \frac{3L}{2}c_g + 8Lc_0 + 3L$. The B_μ^2 term satisfies $\eta_t E + \frac{3L}{2}\eta_t^2E^2 \leq (1 + \frac{3}{2}c_0)\eta_t E$, giving $c_2 = 2 + 3c_0$. Full derivation in Appendix ??.

Theorem 1 (Surrogate Objective Convergence). *Let $\eta_t = \eta_0/\sqrt{t+1}$ and assume the step-size condition holds. Under Assumptions ??, ??, and ??:*

$$\begin{aligned} \min_{0 \leq t < T} \mathbb{E}\|\nabla F(\omega_t)\|^2 &\leq \frac{8\Delta_F}{E\eta_0\sqrt{T}} + \frac{8\mathcal{R}_\infty}{E\eta_0\sqrt{T}} + 32c_2G^2E \\ &\quad + \frac{8c_1E\eta_0(1 + \ln T)}{\sqrt{T}} (\sigma_l^2 + \kappa^2 + 4G^2), \end{aligned} \quad (8)$$

where $\Delta_F \triangleq F(\omega_0) - F_{\inf}$ and $\mathcal{R}_\infty \triangleq \sum_{t=0}^\infty (\frac{8}{\eta_t E} + \frac{3L}{2}) \mathbb{E}\|r_t\|^2 < \infty$.

Proof sketch. Take full expectation of Eq. (??) and sum over $t = 0$ to $T - 1$ (let $S_T \triangleq \sum_{t=0}^{T-1} \eta_t$). The five terms become: $\mathbb{E}[F(\omega_T)] \leq \mathbb{E}[F(\omega_0)]$ (telescoping cancellation), $-\frac{E}{8} \sum \eta_t \mathbb{E}\|\nabla F\|^2$, $c_1E^2(\sigma_l^2 + \kappa^2 + 4G^2) \sum \eta_t^2$, $4c_2G^2E \cdot S_T$, and $\mathcal{R}_\infty$.

Using $F(\omega_T) \geq F_{\inf}$ and rearranging: $\frac{E}{8} \sum \eta_t \mathbb{E}\|\nabla F\|^2 \leq \Delta_F + c_1E^2(\dots) \sum \eta_t^2 + 4c_2G^2E \cdot S_T + \mathcal{R}_\infty$. Extracting the minimum: $\min_{0 \leq t < T} \leq \frac{8}{ES_T} [\Delta_F + c_1E^2(\dots) \sum \eta_t^2 + 4c_2G^2E \cdot S_T + \mathcal{R}_\infty]$.

With $\eta_t = \eta_0/\sqrt{t+1}$: $S_T \geq \eta_0\sqrt{T}$ and $\sum \eta_t^2 \leq \eta_0^2(1 + \ln T)$. The four upper bounds are: initial gap $\leq \frac{8\Delta_F}{E\eta_0\sqrt{T}}$, noise term $\leq \frac{8c_1E\eta_0(1+\ln T)}{\sqrt{T}}(\sigma_l^2 + \kappa^2 + 4G^2)$, bias term $\frac{8}{ES_T} \cdot 4c_2G^2E \cdot S_T = 32c_2G^2E$ (error floor, S_T cancels), and residual $\leq \frac{8\mathcal{R}_\infty}{E\eta_0\sqrt{T}}$. $\mathcal{R}_\infty \leq D \sum (t+1)^{-3/2-2\rho} < \infty$ (since $\rho > 0$). Full details in Appendix ??.

Corollary 1 (Convergence to Error Neighborhood).

$$\limsup_{T \rightarrow \infty} \min_{0 \leq t < T} \mathbb{E}\|\nabla F(\omega_t)\|^2 = 32c_2G^2E.$$

E. Extension to the Adaptive Objective

Assumption 4 (Regularity and Drift). $\tilde{F}_{k,t}$ is L -smooth with the same heterogeneity bound as Assumption ??; $g_{t,e}^k$ is unbiased for $\nabla \tilde{F}_{k,t}(\omega_t)$. There exist ε_t such that $\sup_\omega |\tilde{F}_{t+1}(\omega) - \tilde{F}_t(\omega)| \leq \varepsilon_t$ and $\sum_{t=0}^\infty \varepsilon_t \leq \mathcal{E}_\infty < \infty$.

Proposition 4 (Sufficient Drift Condition). *Let $z_t = (\{\lambda_t^k\}, \mu_t, \omega_t, \omega_t^{\text{proxy}}, \{W_{r,t}^k\})$ collect the adaptive quantities. If the total variation is finite ($\sum_{t=0}^\infty \|z_{t+1} - z_t\| < \infty$), then the drift condition in Assumption ?? holds with $\varepsilon_t = L_z\|z_{t+1} - z_t\|$, yielding $\mathcal{E}_\infty < \infty$. This is typically satisfied in practice due to decaying adaptive schedules.*

Proposition 5 (Lemma Transfer). *Under the above conditions, Lemmas ??-?? remain valid when $F_k, F, \nabla F$ are replaced by $\tilde{F}_{k,t}, \tilde{F}_t, \nabla \tilde{F}_t$. Key insight: inside $\mathbb{E}_t$, the adaptive parameters are frozen, making $\tilde{F}_t$ behave as a fixed objective; the proofs depend only on the unified constants L, σ_l, κ, M .*

Theorem 2 (Adaptive Objective Convergence). *With the same η_t and step-size condition, under Assumptions ??, ??, and ??:*

$$\begin{aligned} \min_{0 \leq t < T} \mathbb{E}\|\nabla \tilde{F}_t(\omega_t)\|^2 &\leq \frac{8\tilde{\Delta}_0}{E\eta_0\sqrt{T}} + \frac{8\mathcal{E}_\infty}{E\eta_0\sqrt{T}} + \frac{8\mathcal{R}_\infty}{E\eta_0\sqrt{T}} \\ &\quad + 32c_2G^2E \\ &\quad + \frac{8c_1E\eta_0(1 + \ln T)}{\sqrt{T}} (\sigma_l^2 + \kappa^2 + 4G^2), \end{aligned} \quad (9)$$

where $\tilde{\Delta}_0 \triangleq \tilde{F}_0(\omega_0) - \tilde{F}_{\text{inf}}$.

Proof sketch. By Proposition ??, Eq. (??) holds for $\tilde{F}_t$. Using $\tilde{F}_{t+1}(\omega_{t+1}) \leq \tilde{F}_t(\omega_{t+1}) + \varepsilon_t$ from Assumption ??, telescoping summation introduces $\sum \varepsilon_t \leq \mathcal{E}_\infty$, which contributes $\frac{8\mathcal{E}_\infty}{E\eta_0\sqrt{T}}$ after division by S_T —decaying at $\mathcal{O}(1/\sqrt{T})$. The error floor $32c_2G^2E$ remains unchanged. Full details in Appendix ??. $\square$

Corollary 2 (Adaptive Error Neighborhood). $\limsup_{T \rightarrow \infty} \min_{0 \leq t < T} \mathbb{E} \|\nabla \tilde{F}_t(\omega_t)\|^2 = 32c_2G^2E$.

F. Conclusion and Rate Decomposition

Physical origin of the error floor. The error floor $32c_2G^2E$ arises from the non-decaying constant bound $B_\mu^2 = 4G^2$ on the systematic bias m_t in $b_t = m_t + \zeta_t$. During telescoping summation, the $\eta_t E$ contributions from m_t accumulate proportionally to S_T ; dividing by S_T cancels the dependence, leaving a nonzero constant. G (Assumption ??) is the maximum client gradient norm, analogous to a fixed noise floor in communication systems, determining the ultimate accuracy ceiling.

Rate decomposition. The upper bound in Theorem ?? consists of five components listed in Table ??. As $T \rightarrow \infty$, all decaying terms vanish, and the squared gradient norm approaches the neighborhood of $32c_2G^2E$.

TABLE I
CONVERGENCE RATE DECOMPOSITION

Term	Rate
$\frac{8\tilde{\Delta}_0}{E\eta_0\sqrt{T}}$ (initial potential)	$\mathcal{O}(1/\sqrt{T})$
$\frac{8\mathcal{E}_\infty}{E\eta_0\sqrt{T}}$ (objective drift)	$\mathcal{O}(1/\sqrt{T})$
$\frac{8\mathcal{R}_\infty}{E\eta_0\sqrt{T}}$ (proxy residual)	$\mathcal{O}(1/\sqrt{T})$
$32c_2G^2E$ (error floor)	non-decaying
$\frac{8c_1E\eta_0(1+\ln T)}{\sqrt{T}}(\dots)$	$\mathcal{O}(\ln T/\sqrt{T})$

APPENDIX A COMPLETE DERIVATIONS OF LEMMAS

Lemma ?? (Client Drift Bound)

Step 1 — Drift recursion. $\omega_{t,e}^k - \omega_t = -\eta_t \sum_{i=0}^{e-1} g_{t,i}^k$. Cauchy–Schwarz: $\|\omega_{t,e}^k - \omega_t\|^2 \leq \eta_t^2 e \sum_{i=0}^{e-1} \|g_{t,i}^k\|^2$. Taking $\mathbb{E}_t$ and weighted summation: $D_e \leq \eta_t^2 e \sum_{i=0}^{e-1} \sum_k p_k \mathbb{E}_t \|g_{t,i}^k\|^2$.

Step 2a — Noise decomposition. $g_{t,i}^k = \nabla F_k(\omega_{t,i}^k) + \varepsilon_{t,i}^k$ ($\mathbb{E}_t[\varepsilon] = 0$, $\mathbb{E}_t\|\varepsilon\|^2 \leq \sigma_l^2$). By $\|a+b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$: $\mathbb{E}_t \|g_{t,i}^k\|^2 \leq 2\sigma_l^2 + 2\|\nabla F_k(\omega_{t,i}^k)\|^2$.

Step 2b — Pullback. $\nabla F_k(\omega_{t,i}^k) = \nabla F_k(\omega_t) + (\nabla F_k(\omega_{t,i}^k) - \nabla F_k(\omega_t))$. L -smoothness: $\|\nabla F_k(\omega_{t,i}^k)\|^2 \leq 2\|\nabla F_k(\omega_t)\|^2 + 2L^2\|\omega_{t,i}^k - \omega_t\|^2$.

Step 2c — Combine. $\mathbb{E}_t \|g_{t,i}^k\|^2 \leq 2\sigma_l^2 + 4\|\nabla F_k(\omega_t)\|^2 + 4L^2 \mathbb{E}_t \|\omega_{t,i}^k - \omega_t\|^2$.

Step 2d — Weighted summation. $\sum_k p_k \mathbb{E}_t \|g_{t,i}^k\|^2 \leq 2\sigma_l^2 + 4\sum_k p_k \|\nabla F_k(\omega_t)\|^2 + 4L^2 D_i$. Using the heterogeneity assumption ?? : $\leq A_t + 4L^2 D_i$, $A_t \triangleq 2\sigma_l^2 + 4\kappa^2 + 4M\|\nabla F\|^2$.

Steps 3–5 — Solve via prefix maximum. Substituting yields $D_e \leq \eta_t^2 e^2 A_t + 4L^2 \eta_t^2 e \sum_{i=0}^{e-1} D_i$. Defining $D_{\max,e} \triangleq \max_{0 \leq j \leq e} D_j$: $D_{\max,e} \leq \eta_t^2 e^2 A_t + 4L^2 \eta_t^2 e^2 D_{\max,e}$. Under $8L^2 \eta_t^2 E^2 \leq 1$, we have $4L^2 \eta_t^2 e^2 \leq 1/2$, so $D_{\max,e} \leq 2\eta_t^2 e^2 A_t$. $\square$

Lemma ?? (Gradient Bias and Second Moment)

Bias term δ_t : $\delta_t = \frac{1}{E} \sum_{e,k} p_k (\nabla F_k(\omega_{t,e}^k) - \nabla F_k(\omega_t))$. By Jensen’s inequality and L -smoothness: $\mathbb{E}_t \|\delta_t\|^2 \leq \frac{L^2}{E} \sum_{e=0}^{E-1} D_e \leq L^2 D_{\max,E} \leq 2L^2 \eta_t^2 E^2 A_t$. This establishes Eq. (??).

Second moment $\mathbb{E}_t \|g_t\|^2$: $\mathbb{E}_t \|g_t\|^2 \leq A_t + 4L^2 D_{\max,E} \leq (1 + 8L^2 \eta_t^2 E^2) A_t$. Substituting $A_t = 2\sigma_l^2 + 4\kappa^2 + 4M\|\nabla F\|^2$:

$$(1 + 8L^2 \eta_t^2 E^2) A_t = 2(1 + 8L^2 \eta_t^2 E^2) \sigma_l^2 + 4(1 + 8L^2 \eta_t^2 E^2) \kappa^2 + 4M(1 + 8L^2 \eta_t^2 E^2) \|\nabla F\|^2.$$

Using $\eta_t \leq \bar{\eta}$ and the definition of c_g (Eq. (??)): $\frac{1}{2}c_g \geq 2(1 + 8L^2 \bar{\eta}^2 E^2)$, $c_g \geq 4(1 + 8L^2 \bar{\eta}^2 E^2)$, and $c_g \geq 4M(1 + 8L^2 \bar{\eta}^2 E^2)$. The three terms are bounded by $\frac{c_g}{2} \sigma_l^2$, $c_g \kappa^2$, and $c_g \|\nabla F\|^2$, summing to $\leq c_g(\sigma_l^2 + \kappa^2 + \|\nabla F\|^2)$. $\square$

Lemma ?? (Single-Step Descent)

(I) Inner product. $\mathbb{E}_t \langle \nabla F, \Delta_t \rangle = -\eta_t E \langle \nabla F, \mathbb{E}_t[g_t] \rangle - \eta_t E \langle \nabla F, m_t \rangle + \mathbb{E}_t \langle \nabla F, r_t \rangle$. Using $\mathbb{E}_t[g_t] = \nabla F + \delta_t$:

$$\begin{aligned} \text{(A)} \quad & -\eta_t E \langle \nabla F, \nabla F + \delta_t \rangle = -\eta_t E \|\nabla F\|^2 - \eta_t E \langle \nabla F, \delta_t \rangle, \\ \text{(B)} \quad & -\eta_t E \langle \nabla F, m_t \rangle, \quad \text{(C)} \quad \mathbb{E}_t \langle \nabla F, r_t \rangle. \end{aligned}$$

Applying Young’s inequality (with $\alpha = 1/2$ for (A), $\alpha = 1/2$ for (B), $\alpha = \eta_t E/4$ for (C)):

$$\begin{aligned} \text{(A)} \leq & -\eta_t E \|\nabla F\|^2 + \frac{\eta_t E}{4} \|\nabla F\|^2 \\ & + \eta_t E \|\delta_t\|^2 = -\frac{3\eta_t E}{4} \|\nabla F\|^2 + \eta_t E \|\delta_t\|^2, \\ \text{(B)} \leq & \frac{\eta_t E}{4} \|\nabla F\|^2 + \eta_t E \|m_t\|^2, \\ \text{(C)} \leq & \frac{\eta_t E}{8} \|\nabla F\|^2 \\ & + \frac{2}{\eta_t E} \mathbb{E}_t \|r_t\|^2. \end{aligned}$$

Summing:

$$\begin{aligned} \mathbb{E}_t \langle \nabla F, \Delta_t \rangle \leq & -\frac{3}{8} \eta_t E \|\nabla F\|^2 \\ & + \eta_t E \|\delta_t\|^2 \\ & + \eta_t E \cdot B_\mu^2 + \frac{2}{\eta_t E} \mathbb{E}_t \|r_t\|^2. \end{aligned}$$

(II) Quadratic term. $\|\Delta_t\|^2 \leq 3\eta_t^2 E^2 \|g_t\|^2 + 3\eta_t^2 E^2 \|b_t\|^2 + 3\|r_t\|^2$. Multiplying by $L/2$, taking $\mathbb{E}_t$, and substituting bounds:

$$\begin{aligned} \frac{L}{2} \mathbb{E}_t \|\Delta_t\|^2 \leq & \frac{3L}{2} \eta_t^2 E^2 c_g (\sigma_l^2 + \kappa^2 + \|\nabla F\|^2) \\ & + \frac{3L}{2} \eta_t^2 E^2 (B_\mu^2 + B_V^2) + \frac{3L}{2} \mathbb{E}_t \|r_t\|^2. \end{aligned}$$

(III) Merging. Substituting $\eta_t E \|\delta_t\|^2 \leq 8L^2 \eta_t^3 E^3 (\sigma_l^2 + \kappa^2 + M\|\nabla F\|^2)$ (from Eq. (??)):

$$\begin{aligned} \mathbb{E}_t[F(\omega_{t+1})] \leq & F(\omega_t) - \frac{3}{8} \eta_t E \|\nabla F\|^2 + \frac{3L}{2} \eta_t^2 E^2 c_g \|\nabla F\|^2 \\ & + 8L^2 \eta_t^3 E^3 M \|\nabla F\|^2 \\ & + \left(\frac{3L}{2} c_g \eta_t^2 E^2 + 8L^2 \eta_t^3 E^3 \right) (\sigma_l^2 + \kappa^2) \end{aligned}$$

$$+ \frac{3L}{2} \eta_t^2 E^2 B_V^2 + \eta_t E B_\mu^2 + \frac{3L}{2} \eta_t^2 E^2 B_\mu^2 \\ + \left(\frac{2}{\eta_t E} + \frac{3L}{2} \right) \mathbb{E}_t \|r_t\|^2.$$

(IV) Coefficient absorption. The coefficient of $\|\nabla F\|^2$ is $-\frac{3}{8} \eta_t E + \frac{3L}{2} \eta_t^2 E^2 c_g + 8L^2 \eta_t^3 E^3 M = -\frac{3}{8} \eta_t E + \eta_t E \cdot \eta_t E L \cdot (\frac{3}{2} c_g + 8L \eta_t E M)$. Under $\eta_t E L \leq c_0 \leq \frac{1}{16(1+c_g)}$ and $c_g \geq 4M$, one can verify $\frac{3}{2} c_g + 8L \eta_t E M \leq \frac{7}{2} c_g$, so the coefficient satisfies $\leq -\frac{3}{8} \eta_t E + \eta_t E \cdot c_0 \cdot \frac{7}{2} c_g \leq -\frac{\eta_t E}{8}$ (since $\frac{7}{2} c_0 c_g < \frac{1}{4}$).

(V) Constants c_1, c_2 . Using $8L^2 \eta_t^3 E^3 \leq 8L c_0 \eta_t^2 E^2$ to reduce order: $(\frac{3L}{2} c_g + 8L c_0) \eta_t^2 E^2 (\sigma_l^2 + \kappa^2) + \frac{3L}{2} \eta_t^2 E^2 B_V^2 \leq c_1 \eta_t^2 E^2 (\sigma_l^2 + \kappa^2 + B_V^2)$, with $c_1 = \frac{3L}{2} c_g + 8L c_0 + 3L$. The B_μ^2 term: $\eta_t E + \frac{3L}{2} \eta_t^2 E^2 = \eta_t E + \frac{3}{2} (\eta_t E L) \eta_t E \leq (1 + \frac{3}{2} c_0) \eta_t E$, yielding $c_2 = 2 + 3c_0$. $\square$

APPENDIX B

COMPLETE DERIVATIONS OF THEOREMS

Theorem ?? (Surrogate Objective Convergence)

Step 1 (Constant setup). From Proposition ??: $\|m_t\|^2 \leq B_\mu^2 = 4G^2$, $\mathbb{E}_t \|\zeta_t\|^2 \leq B_V^2 = 4G^2$. From Proposition ??: $\|r_t\|^2 \leq D/(t+1)^{2+2\rho}$.

Step 2 (Telescoping sum). Take full expectation of Eq.(??) and sum. Let $S_T \triangleq \sum_{t=0}^{T-1} \eta_t$. The five terms accumulate as: leftmargin=*,nosep

- $\mathbb{E}_t[F(\omega_{t+1})] \leq \mathbb{E}_t[F(\omega_t)] \implies \mathbb{E}[F(\omega_T)] \leq \mathbb{E}[F(\omega_0)]$ (telescoping cancellation)
- $-\frac{\eta_t E}{8} \|\nabla F(\omega_t)\|^2 \implies -\frac{E}{8} \sum_{t=0}^{T-1} \eta_t \mathbb{E} \|\nabla F(\omega_t)\|^2$
- $c_1 \eta_t^2 E^2 (\sigma_l^2 + \kappa^2 + B_V^2) \implies c_1 E^2 (\sigma_l^2 + \kappa^2 + 4G^2) \sum_{t=0}^{T-1} \eta_t^2$
- $c_2 \eta_t E \cdot 4G^2 \implies 4c_2 G^2 E \cdot S_T$
- $\left(\frac{8}{\eta_t E} + \frac{3L}{2} \right) \mathbb{E}_t \|r_t\|^2 \implies \sum_{t=0}^{T-1} \left(\frac{8}{\eta_t E} + \frac{3L}{2} \right) \mathbb{E} \|r_t\|^2$

Step 3 (Rearranging). From $F(\omega_T) \geq F_{\inf}$: $\frac{E}{8} \sum \eta_t \mathbb{E} \|\nabla F\|^2 \leq \Delta_F + c_1 E^2 (\sigma_l^2 + \kappa^2 + 4G^2) \sum \eta_t^2 + 4c_2 G^2 E \cdot S_T + \mathcal{R}_\infty$.

Steps 4-5 (Extract minimum & series estimates). $\min_{0 \leq t < T} \leq \frac{8}{ES_T} [\Delta_F + c_1 E^2 (\dots) \sum \eta_t^2 + 4c_2 G^2 E \cdot S_T + \mathcal{R}_\infty]$. With $\eta_t = \eta_0 / \sqrt{t+1}$: $S_T \geq \eta_0 \sqrt{T}$ ($T \geq 4$), $\sum \eta_t^2 \leq \eta_0^2 (1 + \ln T)$.

Step 6 (Four-term substitution).

$$\frac{8}{ES_T} \Delta_F \leq \frac{8\Delta_F}{E\eta_0 \sqrt{T}} \\ \frac{8}{ES_T} c_1 E^2 (\dots) \sum \eta_t^2 \leq \frac{8c_1 E \eta_0 (1 + \ln T)}{\sqrt{T}} (\sigma_l^2 + \kappa^2 + 4G^2) \\ \frac{8}{ES_T} \cdot 4c_2 G^2 E \cdot S_T = 32c_2 G^2 E \text{ (error floor, } S_T \text{ cancels)} \\ \frac{8}{ES_T} \mathcal{R}_\infty \leq \frac{8\mathcal{R}_\infty}{E\eta_0 \sqrt{T}}$$

Step 7 (Residual convergence verification). $\mathcal{R}_\infty \leq D \left[\frac{8}{E\eta_0} \sum (t+1)^{-3/2-2\rho} + \frac{3L}{2} \sum (t+1)^{-2-2\rho} \right] < \infty$ (since $\rho > 0$). $\square$

Theorem ?? (Adaptive Objective Convergence)

Step 1. Proposition ?? guarantees that Eq.(??) holds for $\tilde{F}_t$.

Step 2. From $\tilde{F}_{t+1}(\omega_{t+1}) \leq \tilde{F}_t(\omega_{t+1}) + \varepsilon_t$ (Assumption ??), taking full expectation:

$$\mathbb{E}[\tilde{F}_{t+1}] \leq \mathbb{E}[\tilde{F}_t] - \frac{\eta_t E}{8} \mathbb{E} \|\nabla \tilde{F}_t\|^2 + c_1 \eta_t^2 E^2 (\dots) \\ + 4c_2 G^2 \eta_t E + \left(\frac{8}{\eta_t E} + \frac{3L}{2} \right) \mathbb{E} \|r_t\|^2 + \varepsilon_t.$$

Step 3 (Telescoping sum). Summing from $t = 0$ to $T - 1$: $\frac{E}{8} \sum \eta_t \mathbb{E} \|\nabla \tilde{F}_t\|^2 \leq \tilde{\Delta}_0 + \sum \varepsilon_t + c_1 E^2 (\dots) \sum \eta_t^2 + 4c_2 G^2 E \cdot S_T + \mathcal{R}_\infty$. Dividing by S_T and taking the minimum: $\frac{8 \sum \varepsilon_t}{ES_T} \leq \frac{8\mathcal{E}_\infty}{E\eta_0 \sqrt{T}}$; the error floor $32c_2 G^2 E$ remains unchanged. $\square$